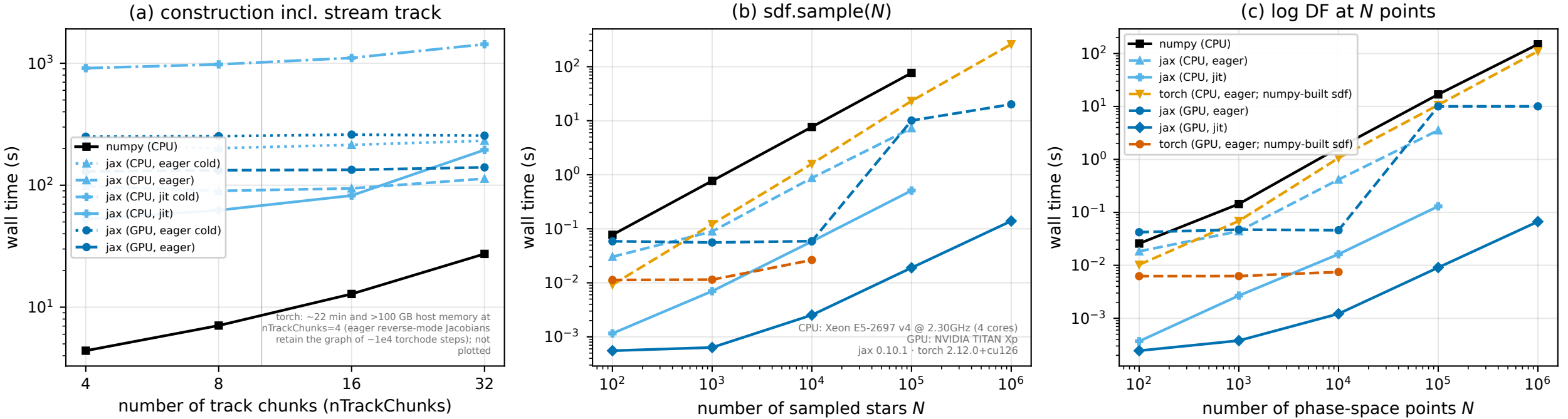
streamdf (action-angle stream model): build once, then sample / evaluate
(a) construction incl. stream track
wall time (s)
number of track chunks (nTrackChunks)
numpy (CPU)
jax (CPU, eager cold)
jax (CPU, eager)
jax (CPU, jit cold)
jax (CPU, jit)
jax (GPU, eager cold)
jax (GPU, eager)
torch: ~22 min and >100 GB host memory at nTrackChunks=4 (eager reverse-mode Jacobians retain the graph of ~1e4 torchode steps); not plotted
(b) sdf.sample(N)
wall time (s)
number of sampled stars N
CPU: Xeon E5-2697 v4 @ 2.30GHz (4 cores)
GPU: NVIDIA TITAN Xp
jax 0.10.1 · torch 2.12.0+cu126
(c) log DF at N points
wall time (s)
number of phase-space points N
numpy (CPU)
jax (CPU, eager)
jax (CPU, jit)
torch (CPU, eager; numpy-built sdf)
jax (GPU, eager)
jax (GPU, jit)
torch (GPU, eager; numpy-built sdf)